Weak convergence of a fast mean-reverting rough Heston model with jumps to a general class of Lévy processes

Abstract

We show weak convergence of the marginals for a re-scaled rough Heston model to a Normal Inverse Gaussian (NIG) Lévy process. This shows we can obtain such a limit without having to impose that the true Hurst exponent H for the model is $\frac{1}{2}$ as in [AC24],[1] or that $H \searrow -\frac{1}{2}$ as in [AAR25], so the result potentially has increased financial relevance. We also extend to the case when V has jumps, where we show weak convergence of the finite-dimensional distributions of the integrated variance to a deterministic time-change of the first passage time process to lower barriers for a more general class of spectrally positive Lévy processes.[2]

1 Asymptotics for the terminal log stock price for a re-scaled rough Heston model

We work on a probability space $(\Omega, \mathcal{F}, \mathbb{Q})$ throughout with filtration $(\mathcal{F}_t)_{t\geq 0}$ satisfying the usual conditions.

Proposition 1.1 *Consider a re-scaled rough Heston model for a log stock price process X_t^ε:*

$$\begin{aligned} dX_t^\varepsilon &= -\frac{1}{2}V_t^\varepsilon dt + \sqrt{V_t^\varepsilon}dB_t, \\ V_t^\varepsilon &= V_0 + \frac{1}{\Gamma(\alpha)}\int_0^t (t-s)^{\alpha-1}\Big(\frac{1}{\varepsilon}\lambda(\theta - V_s^\varepsilon)ds + \frac{1}{\varepsilon}\sigma\sqrt{V_s^\varepsilon}dW_s\Big), \end{aligned} \tag{1}$$

where B and W are two Brownian motions with $dB_tdW_t = \rho dt$ with $\rho \in [-1,0]$, $\alpha \in (\frac{1}{2}, 1)$ and $\lambda, \sigma > 0$. Then (for $t > 0$ fixed) X_t^ε tends weakly to X_t as $\varepsilon \to 0$, where X is a Normal Inverse Gaussian Lévy process which does not depend on $H = \alpha - \frac{1}{2}$.

Proof. Let $I^\alpha(f)(t) = \frac{1}{\Gamma(\alpha)}\int_0^t (t-s)^{\alpha-1}f(s)ds$ denote the αth-order fractional integral of a function f for $\alpha \in (0,1]$, and (without loss of generality), we assume $X_0^\varepsilon = 0$. Then for $p \in (0,1)$ (which will be sufficient for our purposes when we invoke the [Bill86] weak convergence result below)

$$\mathbb{E}[e^{pX_t^\varepsilon}] = e^{V_0 I^{1-\alpha}\phi_\varepsilon(t) + \frac{1}{\varepsilon}\lambda\theta I^1\phi_\varepsilon(t)} \tag{2}$$

where ϕ_ε is the unique solution of the non-linear Volterra integral equation (VIE):

$$\phi_\varepsilon(t) = \frac{1}{\Gamma(\alpha)}\int_0^t (t-s)^{\alpha-1}\Big(\frac{1}{2}(p^2-p) + \frac{1}{\varepsilon}(\rho p\sigma - \lambda)\phi_\varepsilon(s) + \frac{1}{\varepsilon^2}\frac{1}{2}\sigma^2\phi_\varepsilon(s)^2\Big)ds$$

(see e.g. Section 4 in [ER19] or Section 7 in [ALP19]). Now let $\phi_\varepsilon(t) = \varepsilon\psi(\varepsilon^q t)$. Then

$$\begin{aligned} \varepsilon\psi(\varepsilon^q t) &= \frac{1}{\Gamma(\alpha)}\int_0^t (t-s)^{\alpha-1}\Big(\frac{1}{2}(p^2-p) + (\rho p\sigma - \lambda)\psi(\varepsilon^q s) + \frac{1}{2}\sigma^2\psi(\varepsilon^q s)^2\Big)ds \\ &= \frac{1}{\Gamma(\alpha)}\int_0^{\varepsilon^q t} (t-u\varepsilon^{-q})^{\alpha-1}\Big(\frac{1}{2}(p^2-p) + (\rho p\sigma - \lambda)\psi(u) + \frac{1}{2}\sigma^2\psi(u)^2\Big)du\,\varepsilon^{-q} \\ &= \frac{\varepsilon^{-q(\alpha-1)}}{\Gamma(\alpha)}\int_0^{\varepsilon^q t} (\varepsilon^q t-u)^{\alpha-1}\Big(\frac{1}{2}(p^2-p) + (\rho p\sigma - \lambda)\psi(u) + \frac{1}{2}\sigma^2\psi(u)^2\Big)du\,\varepsilon^{-q} \end{aligned}$$

where we set $\varepsilon^q s = u$ in the second line, so $\varepsilon^q ds = du$. Then setting $\varepsilon^q t \mapsto t$, we see that

$$\varepsilon\psi(t) = \frac{\varepsilon^{-q\alpha}}{\Gamma(\alpha)}\int_0^t (t-u)^{\alpha-1}F(\psi(u))du \tag{3}$$

[1] The H parameter in [AC24] is not fixed to be $\frac{1}{2}$, but the Hurst exponent for their family of V^ε processes is of course $\frac{1}{2}$ since the models are all standard Heston for $\varepsilon > 0$.

[2] Note that for citations to results in [ALP19], [BLP24], [BPS24], [FGS21] and [GK19] we are referring to Lemma or Theorem numbers in the published, not the arXiv versions.

where $F(w) = \frac{1}{2}(p^2 - p) + (\rho p\sigma - \lambda)w + \frac{1}{2}\sigma^2 w^2$. If now we let $q = -\frac{1}{\alpha}$, the VIE (3) is independent of ε, so

$$\phi_\varepsilon(t) = \varepsilon\psi(\frac{t}{\varepsilon^{\frac{1}{\alpha}}}).$$

Hence for every $t > 0$ (from Lemma 4.5 in [FGS21])

$$\lim_{\varepsilon\to 0}\frac{1}{\varepsilon}\phi_\varepsilon(t) \quad = \quad \lim_{t\to\infty}\psi(t) \quad = \quad \psi(\infty) \quad = \quad U_1(p) \quad := \quad \frac{1}{\sigma^2}[\lambda - p\sigma\rho - \sqrt{\lambda^2 - 2\lambda\rho\sigma p + \sigma^2 p(1-p\bar{\rho}^2)}]$$

for $p \in (0,1)$, where $\psi(\infty)$ is the smallest root of F and $\bar{\rho}^2 = 1-\rho^2$, and the convergence to $\psi(\infty)$ is rapid for $\varepsilon \ll 1$ since $\frac{1}{\varepsilon}\phi_\varepsilon(t) = \psi(\frac{t}{\varepsilon^{\frac{1}{\alpha}}})$. More precisely, Lemma 4.5 in [FGS21] can be applied to $-\psi$, which solves the following VIE:

$$-\psi(t) = \frac{1}{\Gamma(\alpha)}\int_0^t (t-u)^{\alpha-1}F_1(-\psi(u))du, \quad \text{with } F_1(w) = \frac{1}{2}(p-p^2) + (\rho p\sigma - \lambda)w - \frac{1}{2}\sigma^2 w^2$$

on $\mathbb{R}_+$, since $F_1(0) > 0$ and F_1 is analytic and decreasing on $\mathbb{R}_+$, the aforementioned result in [FGS21] implies that $-\psi(t)$ converges as $t \to \infty$ to the positive root of F_1, which coincides with $-U_1(p)$. Then for the exponent in (2), we know that $\frac{1}{\varepsilon}\phi_\varepsilon(.) = \psi(\frac{t}{\varepsilon^{\frac{1}{\alpha}}})$ is continuous on $\mathbb{R}_+$ and admits finite limit at ∞, and thus is bounded), so (by the dominated convergence theorem) we see that

$$V_0 I^{1-\alpha}\phi_\varepsilon(t) \; + \; \frac{\lambda\theta}{\varepsilon}I^1\phi_\varepsilon(t) \quad = \quad \frac{V_0}{\Gamma(1-\alpha)}\int_0^t (t-s)^{-\alpha}\phi_\varepsilon(s)ds \; + \; \lambda\theta\int_0^t \frac{1}{\varepsilon}\phi_\varepsilon(s)ds \quad \to \quad 0 \; + \; \lambda\theta U_1(p)t$$

as $\varepsilon \to 0$, and $\lambda\theta U_1(p)t$ is the log mgf of a Normal Inverse Gaussian Lévy process (see e.g. Remark 2.3 in [FJ11]). Then from e.g. the solution to Problem 30.4 on Page 573 in [Bill86] (which is also used in [GK19]), X_t^ε tends weakly to the marginal law of an NIG process. ■

1.1 Adding jumps into the V^ε process

If we add a (re-scaled) independent (spectrally positive) compensated jump component into the V^ε process as in Eq 1 in [BPS24] (add more detail on Levy triple), then (from Theorem 3 in [BPS24], see also slide 6 in [Cuch22]), then $F(\psi(u))$ in (3) has an additional term $V_\varepsilon(\varepsilon\psi(u))$, where $V_\varepsilon(p) := \int(e^{pz} - 1 - pz)\varepsilon\nu(\varepsilon z)dz$ is the log mgf of the re-scaled Lévy component, so

$$V_\varepsilon(\varepsilon p) \quad = \quad \int(e^{p\varepsilon z} - 1 - \varepsilon pz)\varepsilon\nu(\varepsilon z)dz \quad = \quad \int(e^{px} - 1 - px)\nu(x)dx \quad = \quad V_1(p)\,. \tag{4}$$

In this case we expect $\frac{1}{\varepsilon}\phi_\varepsilon(t)$ to tend to the smallest root of $F_J(u) = \frac{1}{2}(p^2-p) + (\rho p\sigma - \lambda)u + \frac{1}{2}\sigma^2u^2 + V_1(u)$, assuming a root still exists, and note V_1 (and F_J) are convex (see e.g. Lemma 2.2.5 in [DZ98]).

From here on, we assume $\nu = \nu(z)dz$ only has non-negative support and (state conditions). Then for a spectrally positive Lévy process Z with Lévy triple (b, σ^2, ν) with $b = \rho p\sigma - \lambda$, we know that $\mathbb{E}(e^{uZ_t}) < \infty$ for $u < M$, and (for u in this range) $e^{uZ_t - \Lambda(u)t}$ is an $\mathcal{F}^Z$-martingale.

Let $\Lambda(u) = bu + \frac{1}{2}\sigma^2u^2 + V_1(u)$. Hence from the optional stopping theorem

$$\mathbb{E}(e^{-qH_a}) \quad = \quad e^{-\Phi(q)a} \quad = \quad e^{\Phi(q)|a|} \tag{5}$$

for $q \geq 0$ and $a < 0$ where $H_a = \inf\{t \geq 0 : Z_t < a\}$ and $\Phi(q) \leq 0$ is smallest root of $\Lambda(u) = q$ (since we have a *lower* barrier here), see e.g. Eq 2.5 in [KKR13] for details and the limiting argument using the dominated convergence theorem [3], and (5) also holds for $q < 0$ if q is greater than the minimum value attained by $\Lambda(.)$.

If we now let $W_t^{(-\frac{1}{2})} = W_t^\perp - \frac{1}{2}t$ where $W^\perp$ is another Brownian motion independent of Z and everything else, then for p in some non-zero interval around zero, we see that

$$\mathbb{E}(e^{pW_{H_{-t}}^{(-\frac{1}{2})}}) \quad = \quad \mathbb{E}(e^{\frac{1}{2}(p^2-p)H_{-t}}) \quad = \quad e^{t\Phi_p(-\frac{1}{2}(p^2-p))} \tag{6}$$

and $\Phi_p(-\frac{1}{2}(p^2-p))$ is the smallest root of F_J (note that H_{-t} also depends on p here since $b = \rho p\sigma - \lambda$).

(6) is the mgf of $\eta_{\tau_t^-}$, where $\eta_t = W_t - \frac{1}{2}t$ and $\tau_t^- = \inf\{s : \xi_s < -t\}$, where $\xi_t = -\lambda t + \sigma(\rho W_t + \bar{\rho}\tilde{W}_t) + J_t$, where W and $\tilde{W}$ are independent and J is a spectrally positive Lévy process (independent of $(W, \tilde{W})$), with $\log\mathbb{E}(e^{uJ_1}) = V_1(u)$.

To see this, we note that the joint mgf of (ξ_1, η_1) is

$$\Psi(u,p) \quad = \quad \log\mathbb{E}(e^{u\xi_1 + p\eta_1}) \quad = \quad \Lambda_p(u) + \frac{1}{2}(p^2-p)$$

[3] in [KKR13] they take the largest root since they are considering an upper barrier

with $\Lambda_p(u) = -\lambda u + \frac{1}{2}\sigma^2 u^2 + V_1(u) + \rho\sigma p u$. Now define $\kappa(p)$ by the branch through zero:

$$\Lambda_p(\kappa(p)) + \frac{1}{2}(p^2 - p) \quad = \quad 0$$

with $\kappa(0) = 0$. Then $\kappa(p) = \Phi_p(-\frac{1}{2}(p^2 - p))$, and

$$\mathbb{E}(e^{\kappa(p)\xi_{\tau_t^-} + p\eta_{\tau_t^-}}) \quad = \quad \mathbb{E}(e^{-\kappa(p)t + p\eta_{\tau_t^-}}) \quad = \quad 1$$

so

$$\mathbb{E}(e^{p\eta_{\tau_t^-}}) \quad = \quad e^{\kappa(p)t}$$

assuming $\tau_t^- < \infty$ a.s., which is true if $\Lambda_0'(0+) \leq 0$, where $\Lambda_0(u) = -\lambda u + \frac{1}{2}\sigma^2 u^2 + V_1(u)$ and $V_1'(0) < \infty$.

References

[A21] Abi Jaber, E., "Weak existence and uniqueness for affine stochastic Volterra equations with L^1-kernels", *Bernoulli, 27*(3), 1583-1615, 2021.

[AA25] Abi Jaber, E. and E. Attal, "Simulating integrated Volterra square-root processes and Volterra Heston models via Inverse Gaussian", Preprint, 2025.

[AAR25] Abi Jaber, E., E. Attal and M. Rosenbaum, "From Hyper Roughness to Jumps as $H \to -\frac{1}{2}$", Preprint, 2025.

[ACLP21] Abi Jaber, E., C. Cuchiero, M. Larsson and S. Pulido, "A weak solution theory for stochastic Volterra equations of convolution type", *Annals of Applied Probability, 31*(6), 2924-2952, 2021.

[AC24] Abi Jaber, E. and N. De Carvalho, "Reconciling rough volatility with jumps", *SIAM Journal on Financial Mathematics, 15*(3), 785-823, 2024.

[AE19] Abi Jaber, E. and O. El Euch, "Multifactor approximation of rough volatility models", *SIAM Journal on Financial Mathematics, 10*(2), 309-349, 2019.

[ALP19] Abi Jaber, E., M. Larsson and S. Pulido, "Affine Volterra processes", *Annals of Applied Probability, 29*(5), 3155-3200, 2019.

[AP07] Andersen, L.B.G. and V.V. Piterbarg, "Moment Explosions in Stochastic Volatility Models", *Finance and Stochastics, 11*(1), 29-50, 2007.

[BER25] Ballotta, L., E. Eberlein and G. Rayée, "The term structure of implied correlations between S&P and VIX markets", *Frontiers of Mathematical Finance, 5*, 73-93, 2025.

[Bill86] Billingsley, P., "*Probability and measure*", John Wiley&Sons, 2nd edition, 1986.

[BLP24] Bondi, A., G. Livieri and S. Pulido, "Affine Volterra processes with jumps", *Stochastic Processes and their Applications, 168*, 1042-64, 2024.

[BPS24] Bondi, A., S. Pulido and S. Scotti, "The rough Hawkes Heston stochastic volatility model", *Mathematical Finance, 34*(4), 1197-1241, 2024.

[BG25] Bourgey, F. and J. Gatheral, "The SSR under Quadratic Rough Heston", Preprint, 2025.

[BL24] Boyarchenko, S. and S. Levendorskii, "Correct implied volatility shapes and reliable pricing in the rough Heston model", Preprint, 2024.

[Brez11] Brézis, H., "*Functional analysis, Sobolev spaces and partial differential equations*" (Vol. 2, No. 3), New York: Springer, 2011.

[BF18] Brouste, A. and M. Fukasawa, "Local asymptotic normality property for fractional Gaussian noise under high-frequency observations", *Annals of Statistics, 46*(5), 2045-2061, 2018.

[CD24] Cont, R. and P. Das, "Rough Volatility: Fact or Artefact?", *Sankhya: The Indian Journal of Statistics, Series B, 86*(1), 191-223, 2024.

[CT04] Cont, R. and P. Tankov, "*Financial modelling with Jump Processes*", Chapman&Hall, 2004.

[Cuch22] Cuchiero, C., "Modelling rough covariance processes", Talk, February 2022.

[CT20] Cuchiero, C. and J. Teichmann. "Generalized Feller processes and Markovian lifts of stochastic Volterra processes: the affine case", *Journal of evolution equations, 20*(4), 1301-1348, 2020.

[DZ98] Dembo, A. and O. Zeitouni, "*Large deviations techniques and applications*", Jones and Bartlet publishers, Boston, 1998.

[EFR18] El Euch, O., M. Fukasawa and M. Rosenbaum, "The microstructural foundations of leverage effect and rough volatility", *Finance&Stochastics, 12*(6), 241-280, 2018.

[ER18] El Euch, O. and M. Rosenbaum, "Perfect hedging in rough Heston models", *Annals of Applied Probability, 28*(6), 3813-3856, 2018.

[ER19] El Euch, O. and M. Rosenbaum, "The characteristic function of Rough Heston models", *Mathematical Finance, 29*(1), 3-38, 2019.

[FGS21] Forde, M., S. Gerhold and B. Smith, "Small-time, large-time and $H \to 0$ asymptotics for the rough Heston model", *Mathematical Finance, 31*(1), 203-241, 2021.

[FGS22] Forde, M., S. Gerhold and B. Smith, "Small-time VIX smile and the stationary distribution for the Rough Heston model", Working paper, 2022.

[FJ11] Forde, M. and A. Jacquier, "The large-maturity smile for the Heston model", *Finance and Stochastics, 15*(4), 755-780, 2011.

[FS21] Forde, M and B. Smith, "Rough Heston with jumps - joint calibration to SPX/VIX level and skew as $T \to 0$, and issues with the quadratic rough Heston model", Preprint, 2021.

[FSV21] Forde, M., B. Smith and L. Viitasaari, "Rough volatility and CGMY jumps with a finite history and the Rough Heston model - small-time asymptotics in the $k\sqrt{t}$ regime", *Quantitative Finance, 21*(4), 541-563, 2021.

[FG24] Friz, P. and J. Gatheral, "Computing the SSR", Preprint, 2024.

[GK19] Gatheral, J. and M. Keller-Ressel, "Affine forward variance models", *Finance and Stochastics, 23*, 501-533, 2019.

[GGP19] Gerhold, S., C. Gerstenecker and A. Pinter, "Moment Explosions In The Rough Heston Model", *Decisions in Economics and Finance, 42*(2), 575-608, 2019.

[GLS90] Gripenberg, G., S.O. Londen and O. Staffans, "*Volterra integral and functional equations*" (No. 34), Cambridge University Press, 1990.

[HS25] Han, X. and A. Schied, "The roughness exponent and its model-free estimation", *Annals of Applied Probability, 35*(2), 1049-1082, 2025.

[Jai15] Jaisson, T., "Market impact as anticipation of the order flow imbalance", *Quantitative Finance, 15*(7), 1123-1135, 2015.

[JR16] Jaisson, T. and M. Rosenbaum, "Rough fractional diffusions as scaling limits of nearly unstable heavy tailed Hawkes processes", *Annals of Applied Probability, 26*(5), 2860-2882, 2016.

[JR20] Jusselin, P. and M. Rosenbaum, "No-arbitrage implies power-law market impact and rough volatility", *Mathematical Finance, 30*(4), 1309-1336, 2020.

[KKR13] Kuznetsov, A., A. Kyprianou and V. Rivero, "The theory of scale functions for spectrally negative Lévy processes", Review article, 2013.

[Lew00] Lewis, A.L., "*Option valuation under stochastic volatility*", Finance Press, 2000.

[Prot05] Protter, P. E., "*Stochastic integration and differential equations*", Springer, Berlin, Heidelberg, 2005.

[Rom22] Rømer, S., "Hybrid multifactor scheme for stochastic Volterra equations with completely monotone kernels", Preprint, 2022.

[Sato99] Sato, K., "*Lévy Processes and Infinitely Divisible Distributions*", Cambridge University Press, 1999.

[Sin98] Sin, C., "Complications with Stochastic Volatility Models", *Advances in Applied Probability, 30*, 256-268, 1998.

[Syz23] Szymanski, G., "Optimal Estimation of the Rough Hurst parameter in additive noise", *Electronic Journal of Statistics*, 2023.